Name_____ Date_____ Period_____

LESSON 3-6 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Customary Unit Conversions

Maryland Standard 6.RP.A.3d · Learning target – I can convert between customary units by multiplying or dividing by the conversion rate.

REMEMBER

- 12 in = 1 ft · 3 ft = 1 yd · 16 oz = 1 lb · 2 cups = 1 pt · 2 pt = 1 qt · 4 qt = 1 gal
- From a larger unit to a smaller unit, **multiply**. From a smaller unit to a larger unit, **divide**.
- Check: more of a small unit, fewer of a large unit.

WORKED EXAMPLE

A banner is 54 inches long. How many feet long is it?

Step 1 1 ft = 12 in. Inches to feet is small to large, so divide.

Step 2 54 ÷ 12 = 4.5

Answer **4.5 feet (4 feet 6 inches)**

WORD BANK

convert – write the same amount in a new unit

multiply – use it when the new unit is smaller

divide – use it when the new unit is bigger

facts – 1 ft = 12 in · 1 yd = 3 ft · 1 lb = 16 oz · 1 gal = 4 qt · 1 qt = 2 pt · 1 pt = 2 cups

PRACTICE

show your work · label units

1. Convert each measure.

Step 1 Is the new unit smaller? Multiply by the fact.

Step 2 Is the new unit bigger? Divide by the fact.

a. 8 ft = ____ in

b. 21 ft = ____ yd

c. 5 lb = ____ oz

d. 12 qt = ____ gal

e. 7 pt = ____ cups

f. 40 oz = ____ lb

2. Write <, >, or = in each blank.

a. 3 ft ___ 40 in

b. 5 pt ___ 3 qt

c. 2 lb ___ 30 oz

d. 9 ft ___ 3 yd

3. Complete the ratio table for gallons and quarts.

Gallons	1	3		7.5
Quarts	4		20	

4. Punch Bowl. A punch bowl holds 3 gallons. How many 1-cup servings does it hold? (1 gal = 4 qt, 1 qt = 2 pt, 1 pt = 2 cups)

Step 1 Gallons to quarts: × 4.

Step 2 Quarts to pints: × 2.

Step 3 Pints to cups: × 2.

Show your work.

Answer_____ cups

5. Reasoning. Tasha needs 2 yards of ribbon. She has 70 inches. She says she has enough because 70 is greater than 2. Is she right? Explain.

Frame *Tasha is ___ because 2 yards = ___ inches, and 70 inches is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a container at home labeled in quarts or ounces. Convert its size to another unit.

Name_____ Date_____ Period_____

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Metric Unit Conversions

Maryland Standard 6.RP.A.3d · Learning target – I can convert between metric units by multiplying or dividing by powers of ten.

REMEMBER

- 1 km = 1,000 m · 1 m = 100 cm · 1 cm = 10 mm · 1 kg = 1,000 g · 1 L = 1,000 mL
- Larger unit to smaller unit: **multiply**. Smaller to larger: **divide**. Scale each digit's place value.

WORKED EXAMPLE

A bag of apples has a mass of 2,400 g. What is its mass in kilograms?

Step 1 1 kg = 1,000 g. Grams to kilograms is small to large, so divide.

Step 2 2,400 ÷ 1,000 = 2.4 (each digit has 1/1,000 its value)

Answer **2.4 kg**

WORD BANK

kilo- – 1,000 of the unit

centi- – one hundredth of the unit

milli- – one thousandth of the unit

match units – change to the same unit before you divide

PRACTICE

show your work · label units

1. Convert each measure.

a. 3.5 km = ____ m

b. 85 cm = ____ m

c. 6 L = ____ mL

d. 450 g = ____ kg

2. **Water Station.** A cooler holds 18 L of water. Each cup holds 300 mL. (1 L = 1,000 mL)

Step 1 Change liters to milliliters: × 1,000.

Step 2 Divide by 300 mL for each cup.

a. How many milliliters are in the cooler? _____ mL

b. How many cups can be filled? _____ cups

3. **Relay Race.** Four runners each run 800 m. How many kilometers do they run in all?

Step 1 Multiply 4 × 800 to get meters.

Step 2 Divide by 1,000 to get kilometers.

a. Total meters: _____ m

b. Total kilometers: _____ km

4. Baking. A recipe needs 1.5 kg of flour. Hana has 3 bags with 400 g of flour in each.

Step 1 Find Hana's grams: 3 × 400.

Step 2 Change 1.5 kg to grams: × 1,000.

Step 3 Compare the two amounts.

a. How many grams of flour does Hana have? _____ g

b. Does she have enough? How many grams more or less? _____

5. Find the Error. Ian wrote 4.5 kg = 0.0045 g. Explain his mistake and find the correct mass.

Frame *Ian's mistake is ___. The correct mass is ___ g because ___.*

TEST PRACTICE

choose the best answer

6. A pencil is 16 cm long. How many millimeters long is it?

A 1.6 mm **B** 160 mm **C** 1,600 mm **D** 0.16 mm

7. Which measure is the same as 0.6 L?

A 6 mL **B** 60 mL **C** 600 mL **D** 6,000 mL

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Find a food label with grams or milliliters. Convert the amount to kilograms or liters.